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Research Paper

Non-destructive and rapid measurement of sugar content in growing cane stalks for breeding programmes using visible-near infrared spectroscopy

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The efficient selection of sugarcane varieties with enhanced sugar content requires simple, rapid, accurate and cost-effective assays. The objective of this research was to develop spectroscopic models for non-destructive evaluation of Commercial Cane Sugar (CCS) in growing cane stalks. A portable visible-shortwave near-infrared (Vis/SWNIR) spectrometer with a wavelength range of 570–1031 nm was applied to cane stalks grown under normal field conditions. The CCS models were developed by partial least squares (PLS) regression using spectra sets obtained at three different times (i.e. morning, afternoon and evening), based on both individual spectra and a combined set. During the in-field measurements, it was found that model performance could be affected by varying sample and instrument temperatures, especially for the combined set. The models had coefficients of determination of the prediction set (r^2) of 0.76, 0.76, 0.78 and 0.69 and root mean square errors of prediction (RMSEP) of 1.01, 1.05, 0.99 and 1.17 CCS for the models constructed with the spectra sets of the morning, afternoon, evening and combined, respectively. These results indicate that the CCS models could be used for the monitoring and screening of sugarcane clone selection, except for the model obtained from the combined spectra set, which was influenced by varying instrument temperature. Thus, heat protection for the portable Vis/SWNIR instrument, allowing it to maintain a constant instrument temperature, should be considered if data collection across different periods of the day is necessary.

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1. Introduction

Variety improvement has played an important role in sugar industries to improve sugarcane quality and especially sugar content in cane stalk. Improvement in cane quality has a high economic impact because it can increase sugar products without increasing costs of harvesting, cane transport or milling. Consequently, many countries consider the sugar content to be an important selection target in sugarcane breeding programmes. Sugar content in cane in Thailand and Australia is indicated by a measurement unit called commercial cane sugar (CCS) (Albertson & Grof, 2004). To obtain the highest profitability, sugarcane should be harvested when it reaches its highest CCS value. Thus, the knowledge of CCS accumulation in each sugarcane variety is important for both farmers and sugar mills alike.

Nowadays there are three conventional methods of CCS measurement in breeding programmes: (a) a labour-intensive extraction method for analysis of juice and residual fibre that requires skill, is time-consuming and needs some chemical substances (Albertson & Grof, 2004); (b) an extracting or shredding method for analysis of juice and residual fibre using a Fourier transform near-infrared (FT-NIR) spectrometer (Taira, Ueno, Saengprachatanarug, & Kawamitsu, 2013); (c) the genetic correlation method, which determines the correlation between CCS and physiological observations, such as cane yield (Jackson, 2005). Both methods (a) and (b) require the destruction of the cane stalks during laboratory analysis. Method (c) can be measured without destruction; however, its accuracy is yet to be quantified.

Roach (1989) has described three phases of sugarcane improvement in breeding programmes. The first phase focuses on selection and crossbreeding among clones of the main varieties. The second phase focuses on the development of interspecific hybrids, and the third phase is the exploitation of the interspecific hybrids. These three phases, when combined, commonly take more than ten years. Destructive testing cannot be used during the first and second phases because each clone has a specific characteristic representing each new variety. So, the CCS of each clone is only estimated using the relationship between CCS and physiological characteristics, which provides low accuracy. This limitation affects clone selection in both the first and second phases, which reduces the chances of obtaining a good clone. The selected clones in the second phase would be produced to confirm the exact values of CCS accumulation of each variety in the third phase. Thus, the third phase needs a large experimental area that is able to produce a sufficient number of stalks for monthly cane cutting and CCS measurements. Moreover, due to the requirement of a large experimental area, breeders can select only a few clones for studying in the third phase. All of these constraints lead to a costly and time-consuming process that leads to slow improvement in sugar content.

Besides the knowledge of CCS accumulation throughout the cane growth, breeders also need to follow up on the change in the chemical composition of an individual cane stalk when there is a change in plant growth conditions, such as climate, soil composition or rainfall. However, following up on the CCS for an individual cane stalk cannot be done using destructive methods. Accordingly, a method that can measure the CCS directly in an intact cane stalk without destruction is strongly needed for breeding programmes. A precise, non-destructive method would help breeders unlock the above limitations. Moreover, there will be a possibility to conduct experiments to improve sugar content based on a powerful control system (such as breeding programmes with a small trial size and a control system in chambers).

Presently, near-infrared (NIR) spectroscopy is widely used to determine the quality of agricultural products. It is a rapid and non-destructive method that can be applied without any sample preparation (Jamshidi, Minaei, Mohajerani, & Ghassemian, 2012; Schaare & Fraser, 2000; Sunoj, Igathinathane, & Visvanathan, 2016). NIR spectroscopy has also been used to assess the quality of fruit (Nascimento, Carvalho, Júnior, Pereira, & Teixeira, 2016; Saranwong, Sornsrivichai, & Kawano, 2004). In breeding programmes, Bailleres, Davrieux, and Ham-Pichavant (2002) and Yeh et al. (2005) used the NIR analysis as a tool for the rapid screening of some major wood characteristics in eucalyptus and pine, respectively. Their models allowed them to rapidly screen large breeding populations for a variety of phenotypic traits. Similarly, Shiroma-Kian, Tay, Manrique, Giusti, and Rodriguez-Saona (2008) and Ayvaz et al. (2016) employed Fourier transform infrared spectroscopy (FTIR) to screen potato breeding lines. In the sugar industry, however, such non-destructive methods for variety screening have not been applied. Research has been conducted on using a portable visible-shortwave near-infrared (Vis/SWNIR) instrument to predict pol and fibre content values of cane stalk with promising results (Maraphum et al., 2018; Phuphaphud, Saengprachatanarug, Posom, Maraphum, & Taira, 2019; Taira, Ueno, Saengprachatanarug, et al., 2013). Accordingly, there is a possibility to predict the CCS of growing cane stalk directly in the field, although previous studies have only been carried out in the laboratory.

However, the practicality of using NIR in the field is constrained by the influence of unknown factors on the spectra in the NIR region. According to previous studies, it is well known that NIR spectra can be very sensitive to temperature (Chapanya, Ritthiruangdej, Mueangmontri, Pattamasuwan, & Vanichsriratana, 2019; Thamasopinkul et al., 2017; Xie & Shi, 2013). Wülfert, Kok, and Smilde (1998) found that the isolated molecular features and inter- or intramolecular features, shown from vibrational spectra, were affected by sample temperature. A high sample temperature means that there is a lot of energy stored in a molecule. So, the energy absorption requirement of the molecules in the high temperature sample is less than that of the low temperature sample, despite the same structure and concentration. Many studies recommended that, when using NIR spectra, one should avoid the variation of sample temperature since the weaker hydrogen bonding of water is easily influenced by temperature, which causes changes in model accuracy (Cozzolino et al., 2007; Hageman, Westerhuis, & Smilde, 2005; Wülfert et al., 1998). Nevertheless, it is not easy to maintain samples at a constant temperature for practical use in the field.

In tropical countries, ambient temperatures can vary widely throughout the day. Spectra obtained in the morning

may have different characteristics compared to spectra measured in the afternoon. Accordingly, the objective of this study was to investigate the effect of field temperature on Vis/SWNIR spectra and the predictive ability of calibration models. Commonly, field temperature can easily affect the sample temperature. Thus, the development of a calibration equation that can compensate for a variety of sample temperatures was examined in this study.

2. Materials and methods

2.1. Sample preparation

The cane samples in this study were supplied by the Faculty of Agriculture at Khon Kaen University from the 2018–2019 harvest season. To obtain a robust model to be applied in breeding programmes, three varieties of sugarcane were included in this study. These three varieties consisted of KK3, K88-92 and UT84-12, which are the major commercial cultivars in Thailand. They are very important for breeding in Thailand because, before any new cultivars are released, their important features, such as the CCS level, must be compared to these commercial varieties. The varieties used in this experiment are the hybrid species of tall perennial grasses in the genus *Saccharum* used for sugar production. They have the genomes of varieties planted in many countries such as ROC1 (Taiwan), CP63-588 (USA), SP70-1143 (Brazil), Q76 (Australia) and so on (Shearman et al., 2018).

The field experiments were conducted in north-eastern Thailand (16°28′ N, 102°48′ E, 195 m above sea level), which is in the tropical savannah climate zone. The experiment period covered both winter and summer seasons with temperatures ranging from 18 to 39 °C. The experiments were performed every week once the sugarcane had 8–12 months of growth. To obtain a wide range of CCS values to use in model creation, samples of several varieties and at various maturity stages were included to obtain a robust model. The samples were prepared as follows:

1. Remove all sugarcane leaves.
2. Wipe sugarcane wax off using soft material such as a sponge.
3. Divide part of the stalk into three sub-sections, the bottom, middle and top sections, as shown in Fig. 1. Thus, the predicted value of one stalk was made up of multiple readings at three different positions along the stalk. Each sub-section is approximately 200 mm in length. This length was designed to obtain sufficient sugarcane juice for measurement of the CCS value using a destructive assessment method.
4. Mark each sub-section with the scanning point number.
5. Measure the temperature of the cane surface using an infrared thermometer (BENETECH GM320, Benetech Products Co., Ltd., Samutprakarn, Thailand) at the scanning point before performing spectral collection.
6. Collect spectra repeatedly at three different times: the morning (8:00–10:00 am), the afternoon (12:30–2:30 pm) and the evening (4:00–6:00 pm). The spectra in each stalk were collected at different times to study the influence of different sample temperatures.
7. After finishing the scanning, take all samples to a laboratory for analysis of the CCS using the conventional method, which is elaborated in section 2.3 (Reference analysis).

2.2. Spectral acquisition

A portable visible shortwave near-infrared (Vis/SWNIR) spectrometer (P-TF1, HNK Engineering, Hokkaido, Japan) was used for scanning. A wavelength of 570–1031 nm with 1 nm resolution was applied. Light was supplied by a 25 W halogen lamp transmitted by fibre optics. The reflected light was detected by a silicon array detector. The field-grown cane sample was directly scanned at the cane surface. The integration time was set to 300 ms, which has previously found suitable for the predictions of pol and fibre (Maraphum et al., 2018; Phuphaphud et al., 2019). Prior to scanning the samples, a standard black sheet (Ethylene-vinyl acetate, EVA) and Teflon plate were scanned as reference materials for dark scanning and white scanning, respectively.

The spectrum of each sample was taken by scanning at eight positions (Fig. 1), and an average was computed from the eight successive scans. The first scan was randomly taken at approximately 20 mm under the growing point on the upper part. Three scans were collected at the same height by moving around the cane circumference from the first position at 90°, 180° and 270° (Taira, Ueno, Saengprachatanarug, et al., 2013). Four additional scans on the lower part (points 5 to 8) were collected in the same manner as the upper part.

2.3. Reference analysis

Post-scanning, each whole sample was ground using a shredder machine to produce fibrous cane. Then, the fibrous cane was weighed and squeezed for juice extraction. The fibrous part was taken to determine the fibre content, while the juice was used for the analysis of brix and pol values.

2.3.1. Fibre content analysis

After the juice was extracted, the fibrous cane residue was collected in a cotton cambric bag and washed in an automatic washing machine for 6 min. The fibrous cane was boiled for 30 min to extract soluble solids and washed again. Finally, to achieve constant mass, the samples were dried at 80 °C using an oven. The calculation of fibre content (%fibre) was defined as (Bureau of sugar experiment stations, 2014):

$$\text{Fibre content} = (W_f / W_i) \times 100 \quad (1)$$

where W_i is the initial mass of the fibrous cane. W_f is the final mass after drying.

2.3.2. Brix and pol in juice analysis

After extraction, the juice was immediately stirred and poured onto a refractometer (Nar – 3Ta, ATAGO, Tokyo, Japan) to measure Brix value. The pol value was determined with a

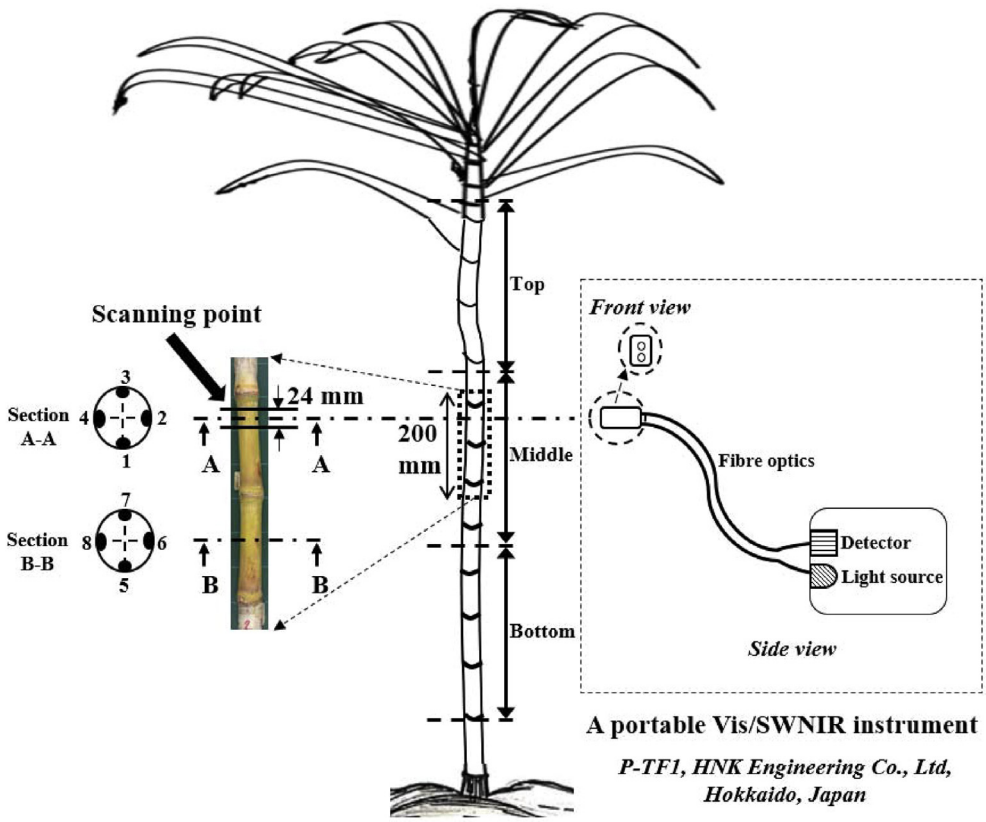

Fig. 1 – Schematic illustration of a portable Vis/SWNIR instrument and measurement positions.

manual polarimeter (Polax – 2L, ATAGO Co., Ltd, Tokyo, Japan) after clarifying the solution with 2 g of calcium hydroxide and 4.66 g of aluminium chloride per 100 ml of sugarcane juice. Pol value (%Pol) was calculated using the following equation (Engelke, 2002):

$$Pol = \{-6.517 + (25.3*OR) - 0.0118*(OR)^2 + (2.937 * Brix) - 0.207*(Brix)^2\} / 100 \quad (2)$$

where OR is the degree of optical rotation measured by the polarimeter, and Brix is the Brix value obtained from the refractometer.

2.3.3. *Determination of the commercial cane sugar (CCS)*

The CCS is a function of Brix in juice, pol in juice and fibre content. It can be defined as follows: (Kirasak, Sansayawichai, Ponragdee, & Thipyawat, 2015):

$$CCS=(0.9433*A*((100-C)/100))-0.5*((0.966*C*((100-C)/100))-(0.9433*A*((100-C)/100))) \quad (3)$$

where A = Brix in juice; B = Pol in juice; and C = Fibre content.

2.4. *Repeatability of reference method and maximum coefficient of determination*

Ten samples were used to measure the CCS values in duplicate using the conventional method and then applied to calculate repeatability (Rep). The Rep value was determined to investigate the precision of the reference methods, which can be defined by the standard deviation of the difference values between duplicates. The Rep value can be applied to estimate the inhomogeneity of the sample. Moreover, it was used to compute the maximum coefficient of determination (R^2_{max}). The R^2_{max} was employed to state the possibility of obtaining the maximum R^2 in the calibration set. The R^2 could be equal to R^2_{max} if there is no error in the spectra (Dardenne, 2010). The R^2_{max} was computed using the following formula:

$$R^2_{max} = \frac{SD_y^2 - Rep^2}{SD_y^2} \quad (4)$$

where SD_y is the standard deviation of the reference data.

2.5. *Spectrum pre-processing*

Before modelling, the spectral data were pre-treated to reduce the effects of baseline shifts and multiplicative scattering that can occur in the case of light scattering or a change in the light-path lengths in the samples. The NIR spectra employed for model development were those with no pre-processing or those pre-processed using the following techniques: baseline offset, range normalisation, mean centring, standard normal variate (SNV), moving average smoothing (MAS), first derivative (D^1) and second derivative (D^2). These pre-treatment techniques have been used to reduce the difference in spectra of the inhomogeneous samples such as shredded cane

(Taira, Ueno, Furukawa, et al., 2013) and wood chips (Posom, Shrestha, Saechua, & Sirisomboon, 2016; Sirisomboon, Funke, & Posom, 2020). All pre-treatments were completed using the MATLAB program (The MathWorks, Inc., Natick, Massachusetts, USA, licence no: 40846673).

2.6. Data analysis

Before modelling, a principal component analysis (PCA) was performed to investigate spectral differences between sugarcane varieties and sample sets (Wang, Wang, Chen, & Han, 2017). In this study, four sets of spectra were used in the modelling. The first three sets were obtained from the scans at different times (morning, afternoon and evening), and the last set was a combination of morning, afternoon and evening scans. From each set 75% was assigned for calibration, whereas 25% was used for validation. The MATLAB program was employed for modelling using an in-house code to perform a partial least square (PLS) regression. The optimal number of latent variables (LV) was considered by a selection of the minimum on the root mean square error of prediction (RMSEP) (Jamshidi et al., 2012). The coefficient of determination of the calibration set (R^2), the root mean square error of calibration (RMSEC), the coefficient of determination of the validation set (r^2), the RMSEP, bias and the ratio of the standard deviation of the reference data to root mean square error of prediction (RPD) were used to evaluate the model performance.

2.7. Reproducibility

To investigate the effects of sample temperature and detector temperature on the Vis/SWNIR spectra, six samples were collected to investigate the reproducibility of the spectra. The reproducibility involved scanning many times by re-loading and re-scanning. Half of the six samples were used for investigating the effect of different sample temperatures, while the remaining samples were used to examine the effect of different detector temperatures.

2.7.1. Reproducibility of different sample temperatures

The sample was put in a resealable plastic bag and heated using a laboratory water bath. The sample temperature was set to one of three levels, 25, 35 and 45 °C, measured by an infrared thermometer. The samples were re-loaded and re-scanned 10 times at the first scanning point (see Fig. 1). Subsequently, the spectra were divided into four sets. Three sets were obtained from the sample scans at each temperature level. Another set consisted of three spectra, including the average spectra of the sample temperature levels of 25, 35 and 45 °C. This set was called the combined set.

2.7.2. Reproducibility of different detector temperatures

In this experiment, the detector temperature of the portable Vis/SWNIR was set to five different levels using spotlights to simulate practical field conditions. The detector temperatures were measured using an infrared thermometer, and the target temperatures were 20, 25, 30, 35 and 40 °C. When the detector temperature reached the target level, the sample was re-loaded and re-scanned 10 separate times (Posom & Sirisomboon, 2017). The spectra used for evaluating the reproducibility were split into six spectra sets. Five sets were from the scanning at each detector temperature level. Another set was a combined set.

2.7.3. Root mean square (RMS) value

The reproducibility was estimated by the root mean square (RMS) value (Xue, Yang, Han, & Chen, 2014):

$$\mathrm{RMS}_j = \sqrt{\frac{\sum_{i=1}^{n}(Yij - \overline{Y}i)^2}{n}} \tag{5}$$

where Y_{ij} is the absorption value of spectrum j at wavelength i, Y_i is the mean absorption value at wavelength i, and n is the number of data points (number of wavelengths). Then, all RMS_j values from each set of spectra were averaged and expressed as RMS values.

2.7.4. Statistical analysis

To investigate whether the spectra pre-processing methods can eliminate the effect of sample temperature or detector temperature, the raw spectra and pre-treated spectra were used to calculate the RMS value. Subsequently, to test the influence of sample temperature and detector temperature on the spectral reproducibility, an analysis of variance (ANOVA) procedure was applied to the RMS values. In this study, the randomised design (CRD) was used as an experimental design to compare the differences between the temperature levels. Duncan's multiple range test (DMRT) was chosen as a method of making multiple comparisons with a significant difference at 5%. The ANOVA and DMRT were done by using IBM SPSS Statistics 26 (SPSS Inc., Chicago, USA, Licence no: 10.101.102.100).

3. Results and discussion

3.1. Overview of optical measurements of cane stalk and statistical characteristics of sugar content

The second derivative spectra of a cane sample, scanned at different sample temperatures, are presented in Fig. 2. A strong absorption band was observed at approximately 966 nm, related to the O–H stretching mode (Vanoli et al., 2012). A slight shift in peak intensity was observed, which tended to decrease with temperature. This result was similar to that obtained by Kawano, Abe, and Iwamoto (1995), who used a Vis/SWNIR spectrometer to determine Brix value in intact peaches. The effect was explained by the weaker hydrogen bonding of water species with increasing temperatures (Thamasopinkul et al., 2017). This temperature effect might influence model accuracy, as the sample temperatures for practical use in the field will vary. The spectra of short-wave bands were used in this research because of their higher energy and penetration depth compared to long-wave bands. Thus, the spectra of short-wave bands can be used to get an adequate sample of inhomogeneous material (Ozanich, 1999), even though the sugarcane surface is thick and hard. The visible region (571–699 nm) was excluded from the model to avoid the influence of cane stalk skin colour on

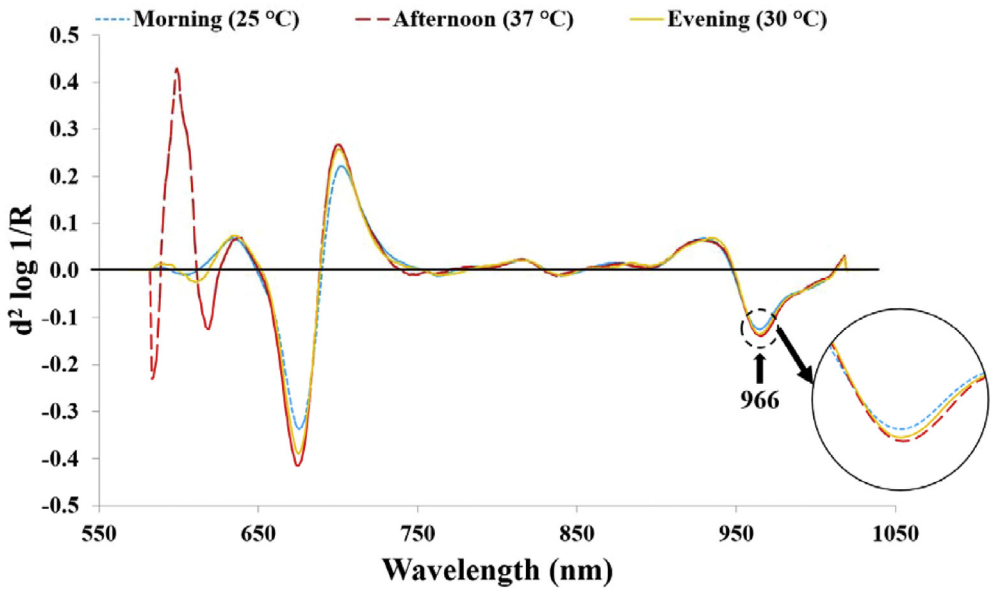

Fig. 2 – Difference spectra of cane stalk, calculated from the second derivative spectra.

measurement sensitivity. Besides the influence of skin colours, using spectra of short-wave bands might result in a low signal-to-noise ratio, especially at the first and last spectra (Liu, He, & Wang, 2008). Thus, the wavelength range between 700 and 1000 nm was considered as suitable for modelling (Phuphaphud et al., 2019).

Figure 3 shows the distribution of sample temperatures obtained in the morning (8:00–10:00 am), afternoon (12:30–2:30 pm) and evening (4:00–6:00pm). The average sample temperature ± SD collected from the morning, afternoon and evening was 25.3 ± 4.6, 36.4 ± 2.6 and 30.0 ± 2.3 °C, respectively. The highest average sample temperature was measured in the afternoon, while the morning had the greatest distribution of sample temperatures. In the evening, the samples had the lowest SD, and the average sample temperature was not high. Therefore, the best data for modelling can be obtained from spectra collected in the evening.

Table 1 shows the statistical data and precision of the reference method. The CCS values for all samples used to develop and validate models had a wide range from 6.44 to 16.77 CCS because samples from several stages of maturity and various sugarcane cultivars were used. The range, average and SD of the CCS values were similar in both calibration and validation sets, due to the systematic sampling. Moreover, to obtain robust models, a wider range of CCS data was used for the calibration set compared to the validation set. Cane varieties in the calibration set included KK3 (35 samples), K88-92 (29 samples) and UT84-12 (37 samples) whereas the validation set consisted of KK3 (10 samples), K88-92 (16 samples) and UT84-12 (8 samples). Figure 4 shows the distribution of CCS values of three sugarcane varieties, indicating that the CCS values of each cultivar had a wide distribution. The repeatability and the R^2_{max} of the reference method were 0.179 and 0.993, respectively. These results demonstrate high precision with

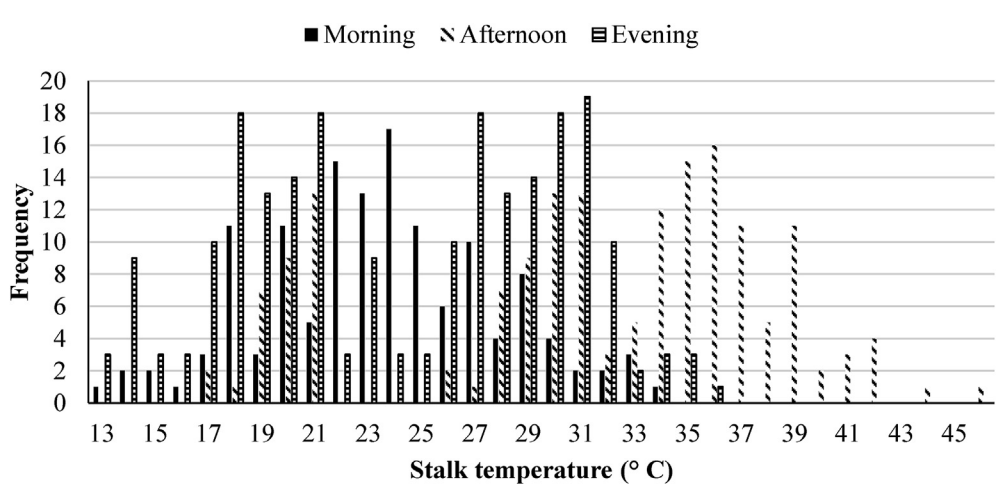

Fig. 3 – Distribution of sample temperatures, measured at different times.

Table 1 – CCS values in cane samples used for model development and validation.

Parameter	Sample sets	N	Max (CCS)	Mean (CCS)	Min (CCS)	SD (CCS)	Repeatability (CCS)	R^2_{max}
CCS	Calibration set	101	16.77	11.54	6.44	2.04	0.179	0.993
	Validation set	34	16.31	11.55	6.81	2.11		

N: number of samples, max: maximum, mean: average, min: minimum, SD: standard deviation.

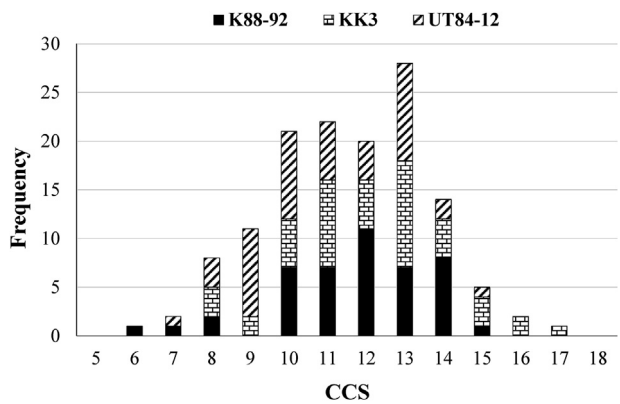

Fig. 4 – Distribution of CCS values of three sugarcane varieties.

the reference method. If there is no error or noise in spectra, the R^2 could be as high as 0.993.

3.2. *PLS models*

The principal component (PC) scores plot in Fig. 5 represents the scatter plot between PC1 and PC2 of three sugarcane cultivars generated from raw spectra. The scatter plot is not categorised into groups, based on both sugarcane varieties and sample sets. These results indicated that having several varieties did not affect the difference in the spectral characteristics between each variety and sample set. Table 2 shows the results of the optimum PLS models for CCS prediction using four sets of spectra, including the morning, afternoon, evening and combined sets. The integration time of 300 ms was applied following the recommendation of Maraphum et al. (2018) and Phuphaphud et al. (2019), who predicted pol and fibre content in cane stalks using a portable visible-shortwave near-infrared (Vis/SWNIR) spectrometer in the laboratory. The latent variables used for model development were 8–10 factors, which were selected based on the lowest RMSEP. The pre-processing technique of derivatives was suitable for all spectra sets, which provided the lowest RMSEP

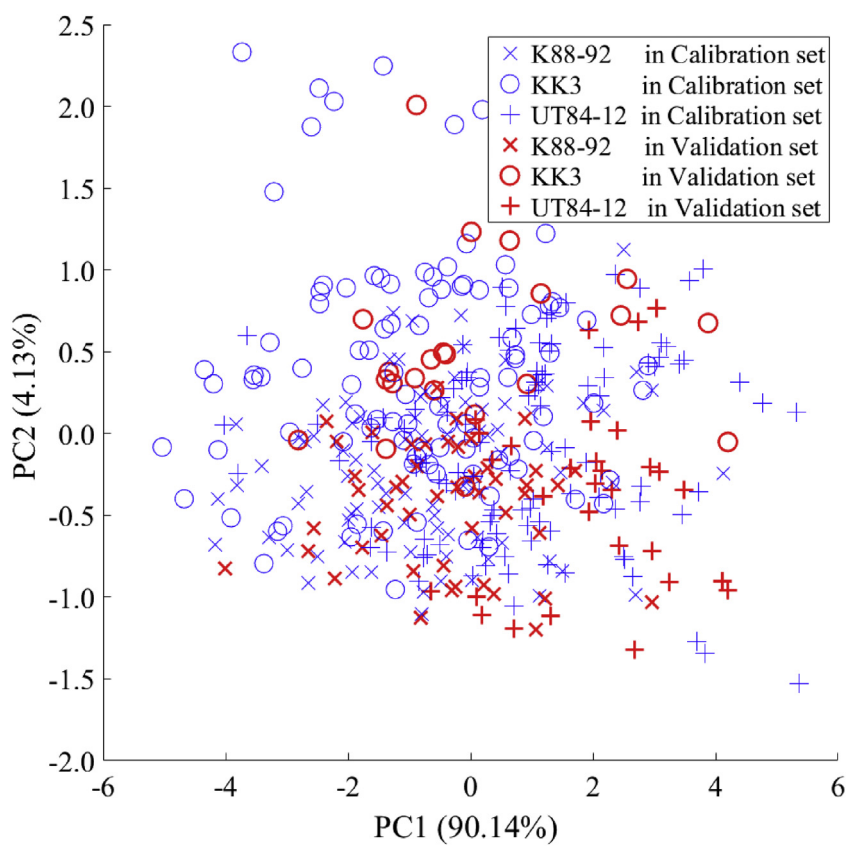

Fig. 5 – PC scores of different sugarcane varieties and different sample sets.

Table 2 – The prediction results of CCS developed by PLS regression using spectra set obtained at three different times, based on both individual and combined spectra sets.

Spectra sets	Pre-processing	Calibration set			Validation set			
		LV	R^2	RMSEC	r^2	RMSEP	bias	RPD
Morning	D^1 (gab = 21, segment = 3)	8	0.76	1.00	0.76	1.01	0.048	2
Afternoon	D^2 (gab = 5, segment = 3)	10	0.81	0.89	0.76	1.05	0.008	2
Evening	D^2 (gab = 7, segment = 3)	10	0.82	0.87	0.78	0.99	−0.019	2
Combined	D^2 (gab = 5, segment = 3)	8	0.70	1.12	0.69	1.17	−0.054	1

LV: latent variables.
R^2: coefficients of determination of calibration set.
r^2: coefficients of determination of prediction set.
RMSEC: the root mean square error of calibration.
RMSEP: the root mean square error of prediction.
Bias: the average of difference between measured values and predicted values.
RPD: ratio of the standard deviation of the reference data to root mean square error of prediction.
D^1: first derivative.
D^2: second derivative.

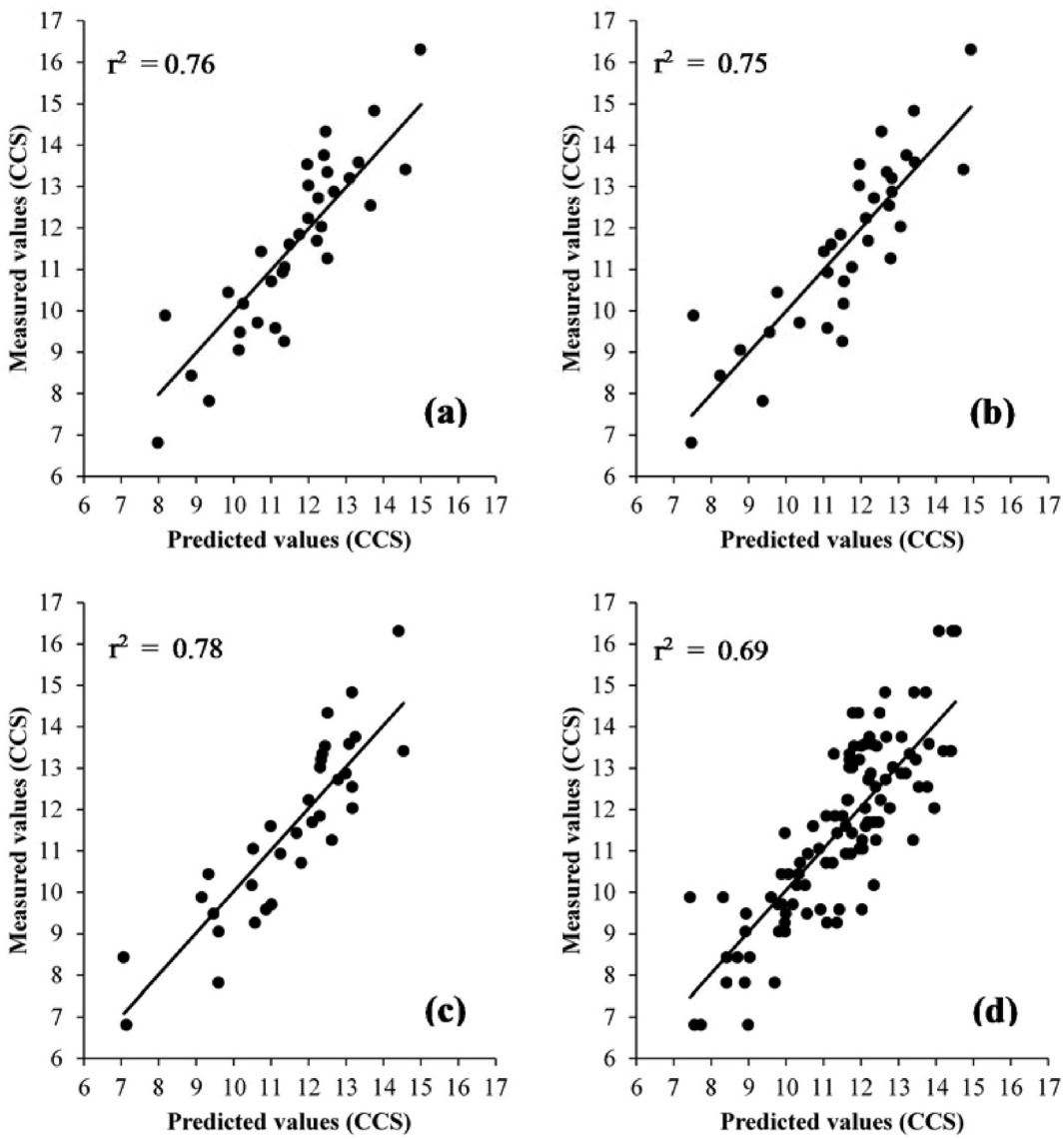

Fig. 6 – Measured versus predicted values for CCS prediction of cane stalks using (a) spectra set of morning, (b) spectra set of afternoon, (c) spectra set of evening, and (d) combined spectra set.

and the highest r^2. The models calibrated with the spectra sets of the morning, afternoon and evening provided a slightly different r^2 of 0.76, 0.75 and 0.78, respectively, while the spectra of the combined set gave the lowest r^2 of 0.69. The scatter plots for the comparison of CCS values predicted by NIR spectroscopy to the values measured by a reference laboratory are shown in Fig. 6. The performance of these models was similar to that found by Nawi, Jensen, and Chen (2012) ($r^2 = 0.70$), who used a Vis/SWNIR spectroradiometer inside a light-proof measurement box to predict the CCS. According to the recommendation of Williams (2007), a model providing an r^2 of 0.66–0.81 could be used for screening and approximate calibrations. So, the models developed in this study could be applied for screening in sugarcane breeding programmes. To examine the effect of sample temperature on model prediction, the scatter plots between residual values and sample temperatures were investigated (see Fig. 7). The residual values were obtained from the measured value minus the predicted value of each sample. Figure 7 shows that the residual does not correlate with the sample temperature, indicating that the derivatives could eliminate the effect of varying sample temperatures.

However, Zornoza et al. (2008) recommended that an approximate prediction is permitted when RPD is in the range of 2.0–2.5, and a poor prediction can be made when RPD is less than 2. This indicates that the model calibrated with the combined spectra (RPD = 1) should be improved. Kawano et al. (1995) developed a calibration equation with temperature compensation for determining the Brix value in intact peaches. Their experiment was performed in a laboratory to control the instrument temperature. Their results show that the model calibrated using spectra measured at 21, 26 and 31 °C can compensate for a variation in sample temperature. Chapanya et al. (2019) developed a temperature compensation model for fermentable sugar prediction in molasses. Consequently, with regard to practical use in the field, the spectra might be strongly influenced by other factors such as sunlight and instrument temperature. As stated by Bornefalk et al. (2013), an increase in the temperature of the silicon detector affects its accuracy by decreasing the detected energy. Thus, it is likely that the instrument temperature is the main factor that affects the accuracy of the NIR spectra.

Figure 8b shows the regression coefficient plot of the best model from Table 2. This plot demonstrates that wavelength plays an important role in each of the predictions. If the coefficient value at any wavelength is high, the vibration of a particular bond at that wavelength has a significant influence on the prediction. Figure 8a illustrates the values obtained by the multiplication of regression coefficient values (see Fig. 8b) and the average spectra of second derivatives spectra (see Fig. 8c) as a function of wavelength. The predicted CCS value is calculated with the sum of the multiplication of regression coefficient values and the average spectra. So, high-intensity peaks at any wavelength mean that those wavelengths in Fig. 8a strongly affect the model prediction.

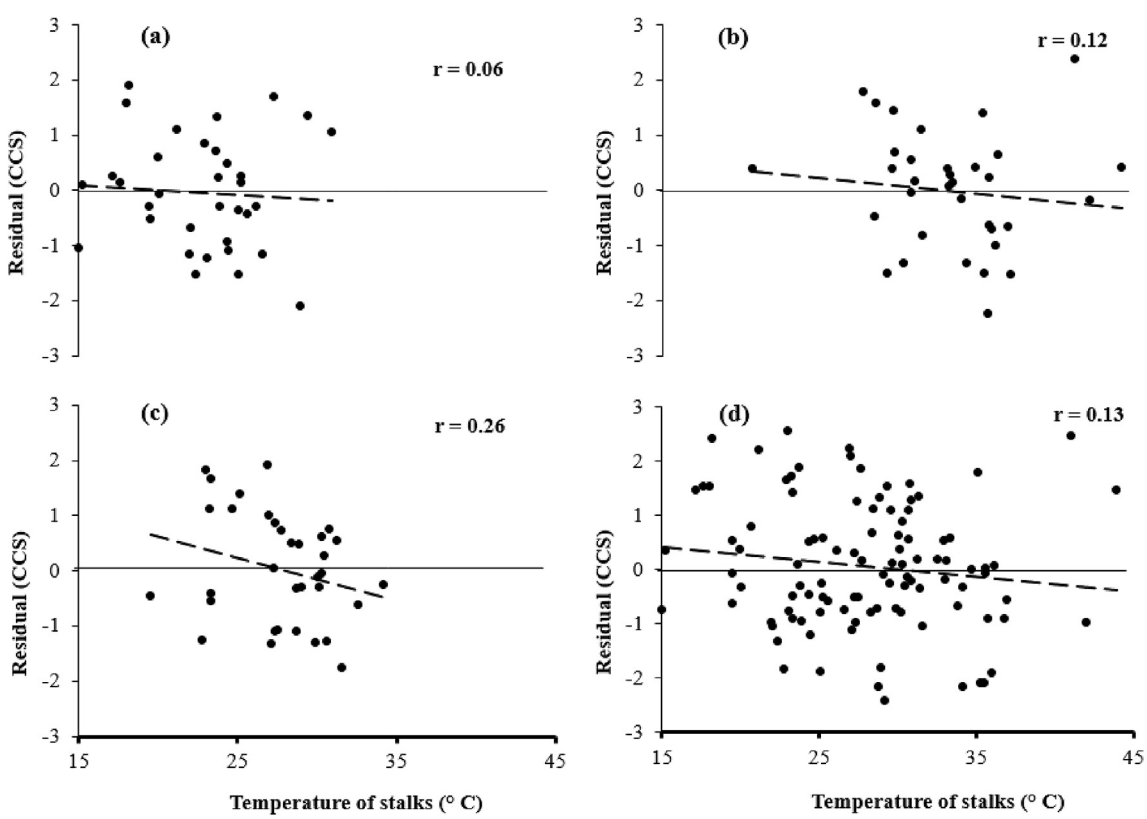

Fig. 7 – Trend of residual values to sample temperatures obtained from (a) morning model, (b) afternoon model, (c) evening model, and (d) combined model.

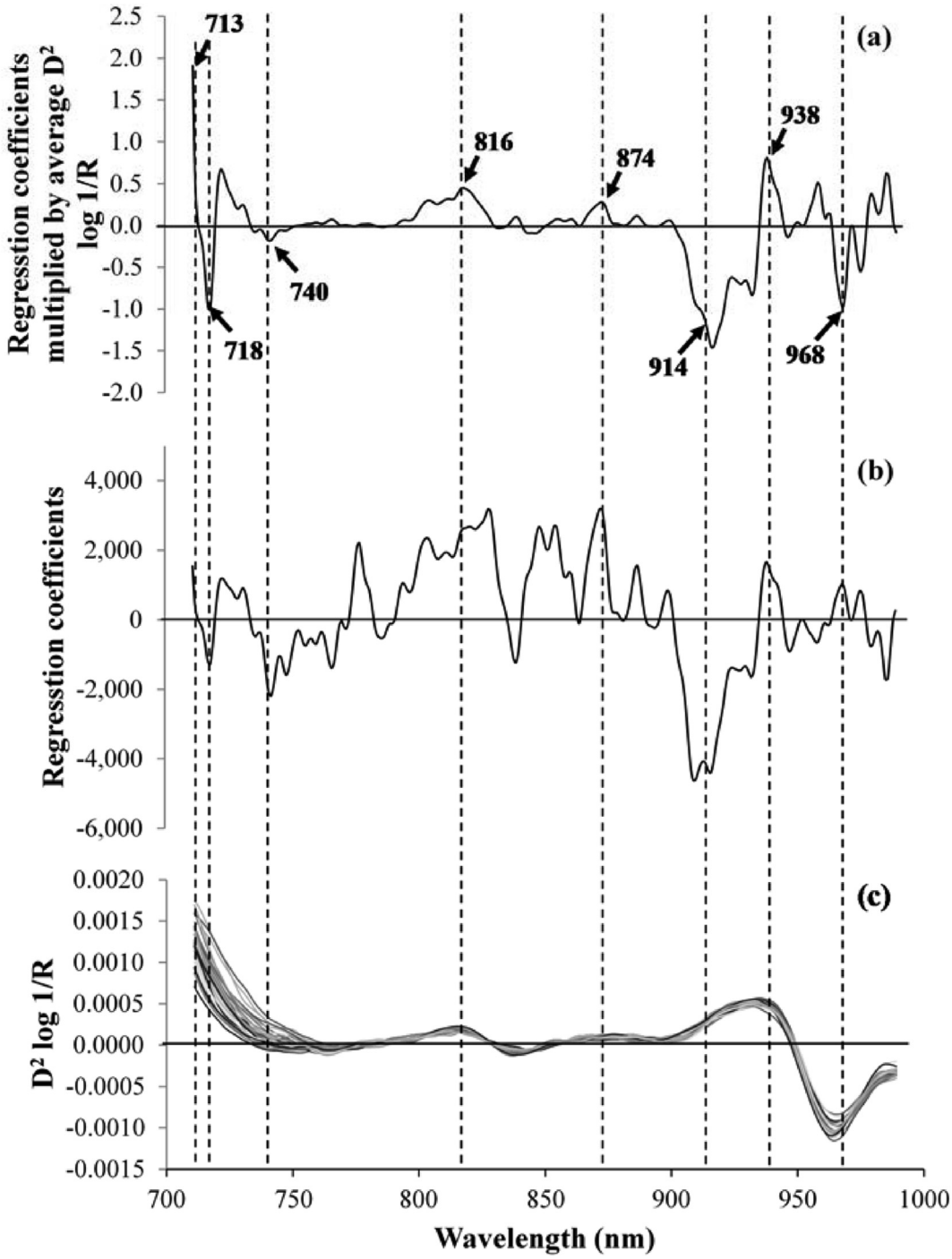

Fig. 8 – Spectra and coefficients on model prediction; (a) multiplication of regression coefficient values and the average spectra of second derivatives spectra; (b) regression coefficient; and (c) second derivatives spectra.

According to the peaks in Fig. 8a, the vibrational bonds of substances related to each wavelength were identified as follows:

1. Peaks at 713, 874 and 938 nm correspond to the third and fourth overtones of the C–H stretching vibration (Workman, Jr. & Weyer, 2007). The wavelength of 938 nm also played an important role in the prediction of the soluble solid content and firmness of European pears, as stated in the study by Wang et al. (2017).
2. The peak at 718 nm is related to water, as found by Chen, Wu, He, and Liu (2011).
3. Peaks at 740 and 816 nm correspond to CH_3 (Workman, Jr. & Weyer, 2007) and cellulose (Williams, 2007) (Williams, 2007), respectively.
4. The peak at 914 nm is correlated to starch (Williams, 2007), as was also found by Saranwong et al. (2004), who analysed starch and dry matter of mango fruit.
5. The peak at 968 nm is related to water (Liu et al., 2008).

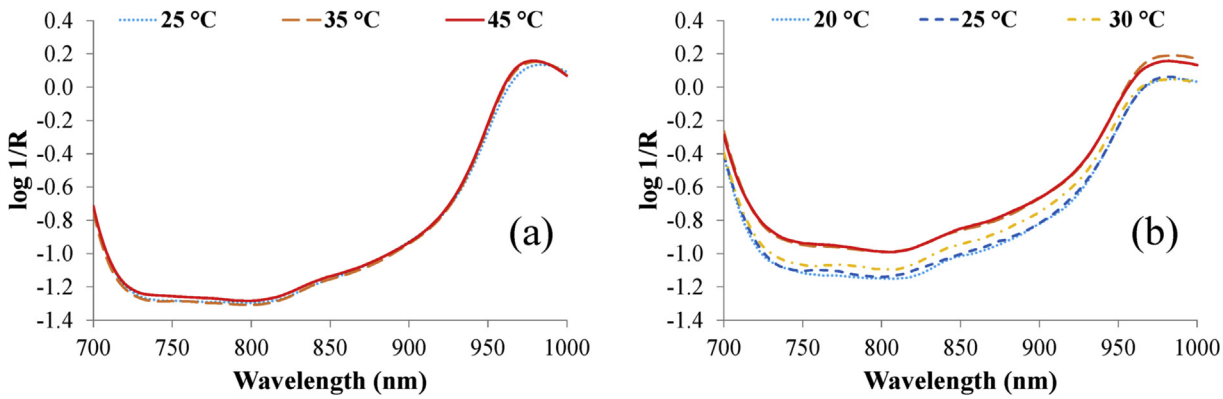

Fig. 9 – Difference in raw spectra of three samples scanned by (a) varying the sample temperatures and (b) varying the detector temperature.

3.3. *The effects of sample temperature and detector temperature on spectral reproducibility*

Figure 9a shows the average spectra of three samples scanned by controlling the instrument temperature at around 25 °C and varying the sample temperatures to three levels (25, 35 and 45 °C), whereas Fig. 9b demonstrates the average spectra of three samples scanned by controlling the sample temperature at around 25 °C and varying the detector temperatures at five levels (20, 25, 30, 35 and 40 °C). Only the shift in the detector temperature leads to noticeable changes in spectra (Fig. 9b).

The ANOVA results (Tables 3 and 4) show no difference in the RMS values for different sample temperatures, while the RMS values for different detector temperatures have significant differences for the combined set. Both raw spectra and pre-treated spectra demonstrate similar results, with the spectra of the combined set in Table 4 provided a much higher RMS value than other sets. These results indicate that detector temperature strongly affected the reproducibility of spectra. This caused reduced accuracy in the model established from the spectra of the combined set.

During practical use in the field, the detector temperature could vary as a result of both ambient temperature change and the radiation from a halogen lamp. An ambient temperature of around 36.4 °C under strong sunlight (clear sky) could increase the detector temperature from 25 to 37 °C when a portable NIR instrument is left for half an hour without scanning, whereas using the portable NIR instrument at room temperature (25 °C) for half an hour could increase the detector temperature from 25 °C to 28 °C. These responses indicate that the main factor increasing detector temperature is the ambient temperature. Figure 10 shows the distribution of ambient temperatures measured while scanning the samples. The ambient temperature varied largely with the time of the day.

Consequently, for practical use in the field, the authors recommend that the range of the detector temperature when taking calibration spectra in the field should be noted and used as a precaution or guideline to ensure that the field measurement is restricted to the same temperature range. This could be planned for in terms of a recommended period of the day in which to obtain measurements. These limitations are proposed to avoid the varying detector temperatures caused by a large range of ambient temperature. But if users wish to use the model that covers all ranges of ambient temperature, it is recommended that the instrument be improved to maintain a constant instrument temperature in the actual field.

This effect of detector temperature on spectral reproducibility could be used as a guideline or precaution when

Table 3 – Results from analysis of variance (ANOVA) tested by Duncan's Multiple Range Test (DMRT) for different sample temperatures.

Spectra pretreatments	Spectra reproducibility (RMS value)				p-value
	25 °C	35 °C	45 °C	Combined	
Raw spectra ^{ns}	0.000652a	0.001379a	0.001131a	0.000208a	0.1468
MSC ^{ns}	0.000232a	0.000182a	0.000276a	0.000105a	0.8507
Baseline Offset ^{ns}	0.000151a	0.000107a	0.000127a	0.000059a	0.6096
Mean Centering ^{ns}	0.000196a	0.000156a	0.000199a	0.000090a	0.7482
Range normalization ^{ns}	0.000437a	0.000771a	0.000562a	0.000089a	0.1888
SNV ^{ns}	0.000627a	0.000444a	0.000529a	0.000248a	0.8514
D^1 ^{ns}	3.96 10^{-7} a	3.29 10^{-7}a	2.85 10^{-7}a	2.02 10^{-7}a	0.9528
D^2 ^{ns}	9.70 10^{-11}a	7.70 10^{-11}a	7.55 10^{-11}a	5.85 10^{-11}a	0.9759

Values with different letters within a row differ significantly ($P < 0.05$).
^{ns} not significant.
* significant.

Table 4 – Results from analysis of variance (ANOVA) tested by Duncan's Multiple Range Test (DMRT) for different detector temperatures.

Spectra pretreatments	Spectra reproducibility (RMS value)						p-value
	20 °C	25 °C	30 °C	35 °C	40 °C	Combined	
Raw spectra*	0.000922a	0.000755a	0.000912a	0.001574a	0.001707a	0.020367b	0.0000
MSC*	0.000068a	0.000026a	0.000040a	0.000025a	0.000029a	0.000450b	0.0000
Baseline Offset*	0.000114a	0.000083a	0.000067a	0.000070a	0.000066a	0.000625b	0.0000
Mean Centring*	0.000098a	0.000064a	0.000056a	0.000054a	0.000052a	0.000592b	0.0000
Range normalization*	0.000236a	0.000109a	0.000152a	0.000099a	0.000118a	0.001734b	0.0000
SNV*	0.000281a	0.000127a	0.000202a	0.000131a	0.000140a	0.002176b	0.0000
D^1*	0.000050a	0.000039a	0.000030a	0.000031a	0.000026a	0.000421b	0.0000
D^2*	0.000017a	0.000012a	0.000008a	0.000012a	0.000008a	0.000111b	0.0000

Values with different letters within a row differ significantly ($P < 0.05$).
^{ns} not significant.
* significant.

calibrating this type of measurement for all cane varieties. The approach from this research might be applied to other countries where they plant similar sugarcane varieties, such as in Australia. Only some exceptions to the physical properties of some varieties should be elucidated, such as length of the internode. However, applying this instrument to other varieties, the model should be revised with added calibration spectra through well-planned practise, with consideration of the detector temperature issues identified here.

3.4. *Comparison of sugarcane CCS accumulation obtained from a portable NIR instrument and conventional method*

Figure 11 shows the change in average CCS values of cane stalk as a function of sugarcane age. The first week was when the sugarcane reached eight months of growth, at which point the sugarcane started to transform starch to sugar, whereas the last week was after the sugarcane reached twelve months of growth at harvest maturity. This figure indicates that the portable NIR instrument can monitor the change of CCS values in each variety, which could help breeders to keep track of the CCS accumulation of the living cane stalks.

According to the results, this research could be applied as a precise screening approach for clone selection in both the first and second phases of the breeding programme instead of the genetic correlation method. It also could help in screening the large population in the third phase and be a helpful method for studying changes in the growing cane's chemical composition when there is a change in plant growth conditions.

Moreover, the particular use of a portable Vis/SWNIR instrument for non-destructive measurement might help unlock the limitations in breeding experiment design, where currently it is only possible to control some environmental variables. This limitation was from having a large number of genotypes and a limit to the number of stalks available for replication in experiments (Gazaffi, Oliveira, Souza, & Garcia, 2014, pp. 203–216). Hence, if the evaluation of sugar content in cane stalks can be done without destructive methods, it would decrease the requirement for replication of each genotype, resulting in a reduction of experimental size and the addition of more environmental variables.

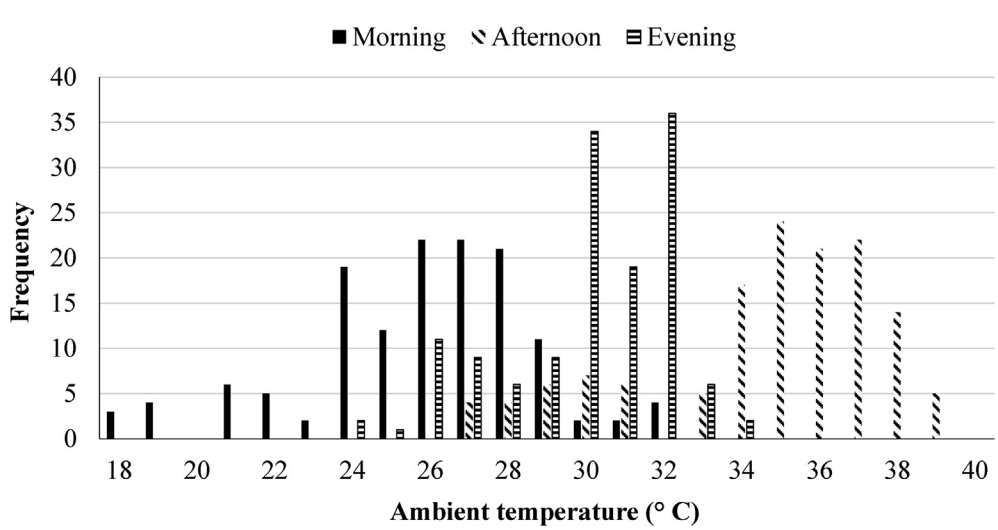

Fig. 10 – Distribution of ambient temperatures, measured while scanning the samples.

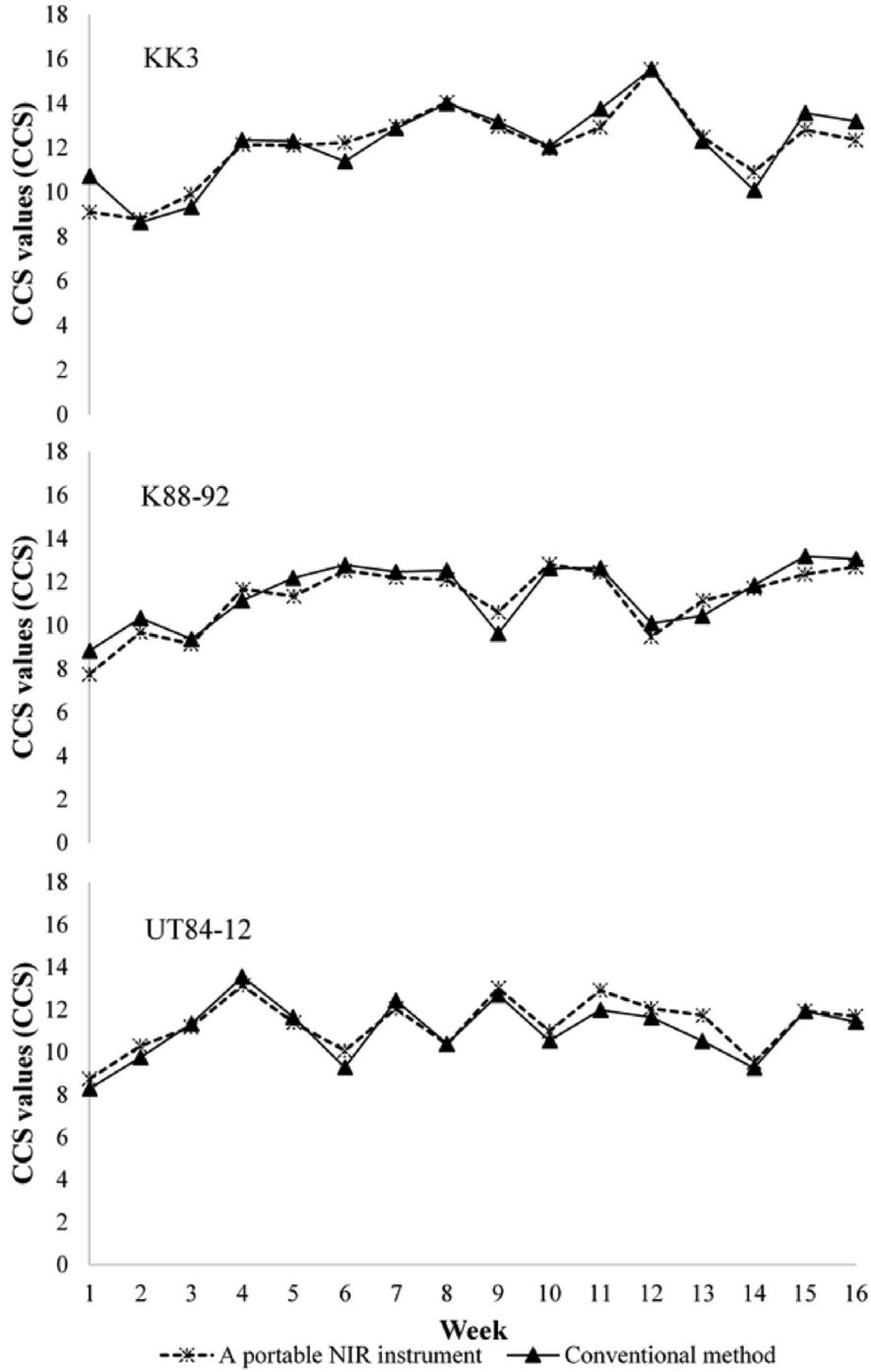

Fig. 11 – Comparison of CCS accumulation of different cane varieties during the 16 weeks maturation stage, obtained from a portable Vis/SWNIR instrument and conventional method.

The portable NIR used in this study had the same structure as the mobile instrument on the market with a specific detector probe arrangement for sugarcane. It also had a similar price. Applying the non-destructive method from this research for CCS monitoring in breeding programmes requires additional costs, but it can provide several advantages: (a) a reduction in the cost of cane transport from field to laboratory, (b) a reduction in the trial size in experiments and an increase in environmental variables, (c) eliminating the need for chemical substances in CCS analysis and (d) avoiding complicated and time-consuming methods of CCS analysis.

4. Conclusions

The results of this study demonstrate that NIR spectroscopy could be used as a non-destructive method for in-field CCS

prediction of growing cane stalks, which will be a helpful tool for monitoring changes in CCS values of growing cane stalks in breeding programmes. Increases in detector temperature led to shifts in spectra, which tended to increase with temperature. The results of the investigation into the effects of sample temperatures and detector temperatures on the spectral reproducibility indicated that various detector temperatures (20–40 °C) greatly affected the reproducibility of spectra, while different sample temperatures (25–45 °C) did not have such influence. The models calibrated with spectra from a specific period of the day provided good accuracy, whereas the model established using spectra sets from all periods could be improved. The spectra collected in the evening period gave the best model with r^2 of 0.78 and RMSEP of 0.99 CCS. Thus, for consistent measurements, any implementation of this data collection method should be performed during the same period of the day as that of the model creation to minimise the difference in ambient temperatures that led to the variation of detector temperatures. To obtain spectra and model that can be used in all ranges of ambient temperature and periods of the day, the instrument should be improved with heat protection to maintain a constant detector temperature when using it in the field.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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